PAGE NO.
ROLL NO.

Practical 3

Aim:- Create custom PHP functions to calculate factorials, primes, etc.

Theory:- In PHP, a function is a block of reusable code designed to perform a specific task. Functions help reduce repetition and make program more structured and easy to understand. Instead of writing the same logic multiple times, a function allows us to define it once and reuse it whenever required.

PHP provides many built-in functions (like strlen(), sqrt(), etc), but developers can also create custom (user-defined) functions to perform specific operations according to their needs.

* Advantages of using functions

(i) Reusability :- Write once, use multiple times

(ii) Modularity :- Break large programs into smaller parts.

(iii) Readability :- Code becomes easier to understand

(iv) Maintenance :- Easy to debug & update.

(v) Efficiency :- Reduces redundancy

* Structure of a PHP Function :-

A function in PHP is defined using the function keyword followed by :-

1. Function Name
2. Parameters (optional)
3. Function body

Syntax:-

```
function functionName ($parameter1, parameter
                                  2) {
        // code to execute
        return value;
  }
```

① Factorial Concept :-

Factorial is a mathematical operation widely used in permulations, combinations and probability.

$n! = n \times (n-1) \times (n-2) \times \ldots \times 1$

Logic used:-

ⓐ Initialize result as 1

ⓑ Multiply numbers from 1 to n using loop

ⓒ Return the result.

② Prime number concept :-

A prime number is a number greater than 1 that is divisible only by 1 and itself.

Logic used:-

ⓐ Check if number is less than or equal to 1

ⓑ Divide number from 2 to $\sqrt{n}$

ⓒ If divisible → not prime

ⓓ If not divisible - prime

* Conclusion:- This practical demonstrate the use of user-defined functions in php to solve mathematical problems efficiency. By using functions, complex logic is simplified into resusable modules, improving program structure, readability, and efficiency.

PAGE NO.
ROLL NO.

Practical 4

* Aim :- Perform string operations (concatenation, slicing)

* Theory :- A string in PHP is a collection or sequence of characters enclosed within single quotes (' ') or double quotes (" "). Strings are one of the most commonly used data types in programming because they follow storage and manipulation of textual information such as names, messages, passwords, address, emails, and user input.

* Single Quotes & double quotes

(i) Single Quotes (' ')
(a) Treat content as plain text
(b) Variables inside single quotes are not interpreted

Ex :-
```
$name = "Rishabh"
echo 'Hello $name';
```
Output :- Hello $name

(ii) Double Quotes (" ")
(a) Variables inside double quotes are interpreted

Ex :-
```
$name = "Rishabh";
echo "Hello $name";
```
Output :- Hello Rishabh

* String Operations :- String operations are action performed on strings to manipulate or analyze text. PHP offers several built-in functions for these operations.

ⓐ String Concatenation :- Concatenation means joining two or more strings into a string. In PHP, concatenation is performed using the dot (.) operator.

Ex:-
```
$str1 = "Hello";
$str2= "World";
echo $str1." ".$str2;
```
Output :- Hello World

ⓑ String Slicing :- String slicing means extracting a portion of a string.
PHP uses the substr() function for slicing

Syntax :- substr(string, start, length);

Example :-
```
$text = "programming";
echo substr($text, 0, 6);
```
Output :- Progra

* Conclusion :- In this practical, string operations concatenation and slicing were performed using PHP. Concatenation helps in combining two or multiple strings into single string and slicing allowed extraction of specific part of a string using the substr() function.